PROBLEM CL2

Bore over a Stagnant Water Layer

Original statement · CL2.pdf

Moving frame | Mass and momentum | Hydrostatic pressure | Froude number

OFFICIAL PROBLEM STATEMENT

The bore sketched in the figure of CL2.pdf moves under the action of gravity with constant velocity U over an initially stagnant water layer of thickness h_0, leaving behind a layer of thickness h_1 moving with velocity U_1. Assuming that in the air both viscous stresses and pressure variations with respect to the ambient value p_a are negligible, and so are viscous stresses on the bottom wall, find the values of U_1 and h_1 as a function of U, h_0 and g. As a preliminary computation, obtain the pressure distributions both upstream and downstream from the bore, verifying that they can be expressed in the form $p = p_a + \rho g(h_0 - z)$ and $p = p_a + \rho g(h_1 - z)$, respectively, where z represents the vertical distance to the bottom wall.

Source: CL2.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Planar flow (per unit width) of an incompressible liquid of density ρ, with gravity $\bar{g} = -g\,\bar{e}_z$ (here gravity is **not** neglected: it drives the bore).
- The bore advances towards $-\bar{e}_x$ with constant velocity U over the stagnant layer h_0; behind it the layer h_1 moves with $-U_1\bar{e}_x$. Far from the front the velocity is uniform and horizontal and the free surface is flat at $p = p_a$.
- Viscous stresses negligible in the water sections and on the bottom wall (given); $\mu/(\rho U h) \ll 1$.
- Data: $[U,\ h_0,\ g,\ \rho]$. Unknowns: $[h_1,\ U_1]$ and the pressure distributions.

Symbol	Meaning	SI unit
U	velocity of the bore	m/s
U_1	velocity of the water behind the bore	m/s
h_0, h_1	layer thickness ahead of and behind the bore	m
$F = U^2/(gh_0)$	Froude number of the bore	1

PHASE 2

Frames, Control Volume and Surfaces

- Laboratory frame L, with z measured from the bottom: unsteady flow, because the front moves.
- Frame of the bore B, moving with $-U\bar{e}_x$ relative to L. It is inertial (constant velocity), so the balances keep their form, and the flow is steady in it. Velocities: $\bar{v}_B = \bar{v}_L + U\bar{e}_x$, so the water ahead has $U\bar{e}_x$ and the water behind $(U - U_1)\bar{e}_x$.
- Control volume fixed in B ($\bar{v}_c = 0$). Surfaces: Σ_0, vertical section ahead ($\bar{n} = -\bar{e}_x$, height h_0); Σ_1, vertical section behind ($\bar{n} = +\bar{e}_x$, height h_1); Σ_b, bottom wall ($\bar{n} = -\bar{e}_z$); Σ_s, free surface (steady material surface in B, $p = p_a$).

PHASE 3

Balances, Surface by Surface

3.1 Preliminary: hydrostatic pressure in the uniform regions

Control volume: a column of width Δx in the uniform region ahead, from height z up to the free surface $z = h_0$. Surfaces: Σ_{bot} (at height z, $\bar{n} = -\bar{e}_z$), Σ_{top} (free surface, $\bar{n} = +\bar{e}_z$) and two vertical faces Σ_{side} ($\bar{n} = \mp\bar{e}_x$). Only the z component is needed.

> Momentum, z component, on the column (Notes Eq. 3.36)
>
> $$\underbrace{\frac{d}{dt}\int_{V_c} \rho v_z \, dV}_{0} + \int_{\Sigma_c} \rho v_z \, (\bar{v}\cdot\bar{n}) \, d\sigma = -\int_{\Sigma_c} p \, n_z \, d\sigma + \underbrace{\int_{\Sigma_c} (\bar{\bar{\tau}}'\cdot\bar{n})_z \, d\sigma}_{\text{negligible}} - \int_{V_c} \rho g \, dV$$
>
> Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,n_z\,d\sigma$
Σ_{bot}	0 $(v_z = 0)$	$+p(z)\,\Delta x$
Σ_{top}	0 $(v_z = 0)$	$-p_a\,\Delta x$
Σ_{side}	0 $(v_z = 0)$	0 $(n_z = 0)$

Volume term: $-\int_{V_c} \rho g\,dV = -\rho g\,(h_0 - z)\,\Delta x$ (weight of the column).

$$0 = p(z)\,\Delta x - p_a\,\Delta x - \rho g\,(h_0 - z)\,\Delta x \implies \boxed{p = p_a + \rho g\,(h_0 - z)} \quad (1)$$

The same column taken behind the bore gives $p = p_a + \rho g\,(h_1 - z)$ (1').

3.2 Mass and momentum in the frame of the bore

`Mass (Notes Eq. 3.11), frame B`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_0	$\rho U\,(-1)\,h_0 = -\rho U h_0$
Σ_1	$\rho\,(U - U_1)\,h_1$
Σ_b	0 (wall)
Σ_s	0 (material surface)

$$\boxed{U\,h_0 = (U - U_1)\,h_1} \quad (2)$$

In the momentum balance p may be replaced by $p - p_a$ because $\int_{\Sigma_c} p_a \bar{n}\, d\sigma = 0$ on a closed surface. The x component is taken (gravity has no x component). On Σ_0 and Σ_1 the pressure is (1) and (1'); the antiderivative of $\rho g(h - z)$ is $\rho g(hz - z^2/2)$, and Barrow's rule between $z = 0$ and $z = h$ gives $\rho g h^2/2$.

> **Momentum, x component (Notes Eq. 3.36), frame B**
>
> $$\underbrace{\frac{d}{dt}\int_{V_c} \rho v_x \, dV}_{0 \text{ (steady in } B)} + \int_{\Sigma_c} \rho v_x \, (\bar{v}\cdot\bar{n})\, d\sigma = -\int_{\Sigma_c} (p - p_a)\, n_x \, d\sigma + \int_{\Sigma_c} (\bar{\bar{\tau}}'\cdot\bar{n})_x \, d\sigma + \underbrace{\int_{V_c} \rho g_x \, dV}_{g_x = 0}$$
>
> Each surface integral, surface by surface (the vertical line drawn under each integral in class):
>
> | **Surface** | $\int \rho v_x\,(\bar{v}\cdot\bar{n})\,d\sigma$ | $-\int (p-p_a)\,n_x\,d\sigma$ | $\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma$ |
> |---|---|---|---|
> | Σ_0 | $\rho U\,(-U)\,h_0 = -\rho U^2 h_0$ | $+\int_0^{h_0} \rho g(h_0 - z)\,dz = \frac{\rho g h_0^2}{2}$ | 0 |
> | Σ_1 | $\rho\,(U - U_1)^2\,h_1$ | $-\int_0^{h_1} \rho g(h_1 - z)\,dz = -\frac{\rho g h_1^2}{2}$ | 0 |
> | Σ_b | 0 | 0 ($n_x = 0$) | 0 (given) |
> | Σ_s | 0 | 0 ($p = p_a$) | 0 (air) |
>
> $$\boxed{-\rho U^2 h_0 + \rho\,(U - U_1)^2\,h_1 = \frac{\rho g}{2}\left(h_0^2 - h_1^2\right)} \qquad (3)$$

3.3 Solution for h_1/h_0 and U_1/U

From (2), $U - U_1 = U h_0/h_1$, so $(U - U_1)^2 h_1 = U^2 h_0^2/h_1$. Substituting into (3) and dividing by ρ:

$$U^2 h_0\left(\frac{h_0}{h_1} - 1\right) = \frac{g}{2}(h_0 - h_1)(h_0 + h_1) \;\Longrightarrow\; U^2 h_0\,\frac{h_0 - h_1}{h_1} = \frac{g}{2}(h_0 - h_1)(h_0 + h_1).$$

For a bore $h_1 \neq h_0$, so $h_0 - h_1$ cancels. With $r = h_1/h_0$ and $F = U^2/(g h_0)$, dividing by $g h_0^2$ gives $F/r = (1 + r)/2$:

$$r^2 + r - 2F = 0 \implies \boxed{\frac{h_1}{h_0} = \frac{1}{2}\left(\sqrt{1+8F} - 1\right)} \qquad (4)$$

(the negative root is rejected, $h_1 > 0$). From (2), $U_1/U = 1 - h_0/h_1 = (r-1)/r$; with $s = \sqrt{1+8F}$, $r = (s-1)/2$ and

$$\boxed{\frac{U_1}{U} = 1 - \frac{2}{s-1} = \frac{\sqrt{1+8F} - 3}{\sqrt{1+8F} - 1}}, \qquad F = \frac{U^2}{g h_0}. \qquad (5)$$

Erratum in the official solution (reading of the handwriting)

CL2.pdf gives $h_1/h_0 = \tfrac{1}{2}(\sqrt{1+8F} - 1)$, as in (4). For the velocity it gives, as legible in the scan, $U_1/U = \tfrac{3}{2}\left(\tfrac{1}{3}\sqrt{1+8F} - 1\right) = (\sqrt{1+8F} - 3)/2 = r - 1$. This contradicts its own continuity equation (2), which requires $U_1/U = (r-1)/r$; the two coincide only for $r = 1$. Check: (2) gives $(U - U_1)h_1 = U h_0 > 0$, hence $U_1 < U$, whereas $r - 1 > 1$ for $F > 3$ (for $F = 4$, $r - 1 = 1.3723$ against $U_1/U = 0.5785$ from (5)). The computed value is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- A bore exists only for $F > 1$: $r > 1 \iff s > 3 \iff F > 1$. F compares the speed of the front with the shallow-water wave speed $\sqrt{g h_0}$.
- Weak bore, $F = 1 + \epsilon$: $r \approx 1 + \tfrac{2}{3}\epsilon$ and $U_1/U \approx \tfrac{2}{3}\epsilon \to 0$ (an infinitesimal gravity wave travelling at $\sqrt{g h_0}$).
- Strong bore, $F \gg 1$: $r \approx \sqrt{2F}$ and $U_1 \to U$.
- Dimensions (SI): both sides of (3) are forces per unit width, $[\rho U^2 h_0] = [\rho g h^2] = \mathrm{N/m}$; F, r and U_1/U are dimensionless.
- Numerical cross-check: $F = 4$, $h_0 = 0.5$ m, $g = 9.81$ m/s^2 give $U = 4.4294$ m/s, $h_1/h_0 = 2.3723$ ($h_1 = 1.1861$ m), $U_1/U = 0.57846$ ($U_1 = 2.5623$ m/s); the residuals of (2) and (3) are below 10^{-14} in SI units.

Extension (beyond the official solution)

Mechanical energy is not conserved across the bore, so Bernoulli's equation cannot be used between both sections. The head lost is the classical hydraulic-jump value $\Delta H = (h_1 - h_0)^3/(4h_0h_1)$, positive for $h_1 > h_0$.